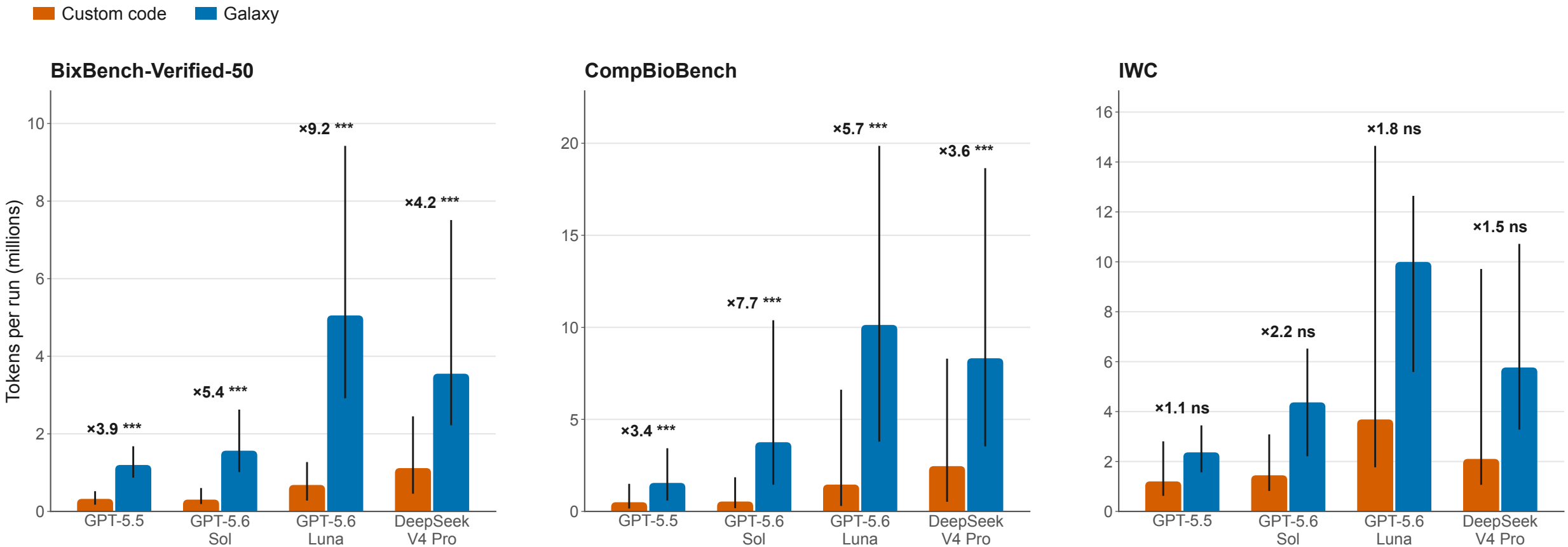
Galaxy runs use 3–9× the tokens of custom code on BixBench and CompBioBench

Bars: median tokens per run (input including cached context, plus output); whiskers: interquartile range. Above each pair: Galaxy ÷ custom code, the geometric mean over tasks of the ratio of their mean tokens (replicates averaged), with the paired cluster sign-flip test, Holm-adjusted over the 12 model comparisons:
* adj. P < 0.05, ** < 0.01, *** < 0.001, ns not significant. Under each panel: the same ratio for the four models together, and the share of input tokens read from the cache.



All four models: ×5.3 (95% CI 4.0 to 7.1), P < 0.001
Cached share of input: custom code 96%, Galaxy 98%

All four models: ×4.8 (95% CI 4.0 to 5.8), P < 0.001
Cached share of input: custom code 97%, Galaxy 97%

All four models: ×1.6 (95% CI 0.8 to 3.2), P = 0.24
Cached share of input: custom code 98%, Galaxy 97%